Network Modeling of Socioeconomic Segregation Across Chicago's Community Areas

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Introduction & Motivation

The aim of our project was to examine the socioeconomic inequalities across Chicago's Community Areas, focusing on the hierarchy between high-income, middle-income, and low-income neighborhoods. We explored these differences through social and economic patterns and spatial segregation, with particular attention to how history, public policies, and uneven investments have shaped the current landscape of inequality in Chicago.

We were motivated to study this topic because Chicago is often celebrated as a vibrant city that many people want to visit, but most public narratives prioritize on a small set of locations such as the iconic downtown and lakefront areas rather than the full city. This selective narrative can hide the very real inequalities that shape the residents in many other neighborhoods.

Our project tries to look past the usual narratives by examining patterns across nine Community Areas that span a hierarchy from wealthy and highly educated to deeply disinvested and with high poverty. By doing so, we aim to understand how opportunities, resources, and living conditions cluster and diverge across the city, and how these inequalities appear when we treat Community Areas as a network of similar or dissimilar places rather than as isolated nodes.

Utilized Data

Our data came from ACS surveys for Chicago between 2018 and 2023. We collected the following data variables for the 9 Community Areas mentioned previously:

1. Race Composition
2. Median Household Income
3. SNAP Reliance
4. Educational Attainment
5. Tenure (Rent vs Ownership)
6. Employment Status
7. Poverty Rate
8. Broadband (Internet) Access

We then combined these datasets into a single aggregated dataset which was used to generate network models and analyze socioeconomic profiles for each Community Area.

Project Design & Execution

We selected nine Community Areas which included three high-income, highly resourceful North Side areas: Lincoln Park; Lake View; Near North Side; three medium-income, mixed profile Northwest areas: Jefferson Park; Portage Park; and Irving Park, and South/West Side areas that are experiencing poverty: Englewood; West Englewood; South Lawndale / Little Village.

Our design had three main layers. First, we constructed a quantitative “persona” for each Community Area based on the census data that included race, income, education, employment status, housing status, governmental monetary assistance, and internet access. Starting from the aggregated dataset described earlier, we converted these variables into a common feature space and normalized them so that large values do not dominate the analysis. From there we were able to produce one standardized feature vector for each of the nine Community Areas that summarizes its socioeconomic persona.

Secondly, we modeled relationships between these profiles as networks. We computed several pairwise similarities and distance matrices on the normalized features which included cosine similarity, Euclidean distance, and correlation. Using the cosine similarity matrix as our main reference, we built an undirected graph in NetworkX where nodes represent Community Areas and edge represent the similarities of the data variables. Additionally, we used a similarity threshold to keep only the strongest ties, which resulted in a sparse, but an interpretable network that highlights which neighborhoods are mostly alike. We also constructed alternative versions of the network by changing the cosine threshold, using k nearest neighbor rules, and switching to Euclidean or correlation-based relationships. These variations allowed us to check how stable the patterns were under reasonable modeling methods.

Lastly, we applied a set of standard network analysis tools to these graphs to study segregation and clustering in the system. On the similarity graphs, we computed degree, eigenvector, and closeness centrality to identify which Community Areas are the core of the network and which are more peripheral. Moreover, we ran Louvain community detection to find groups of neighborhoods that are more similar to each other than to the rest of the graph, and we calculated assortativity by income and education to see how strongly ties tend to connect areas with similar socioeconomic tiers. With additions to that, we measured local and global clustering coefficients to understand the extent to which similar neighborhoods form a group of nodes that are connected. Furthermore, we also applied k means clustering directly to the feature space and visualized the results with principal component analysis, which produced the cluster structure shown in our poster.

All the mentioned steps above were implemented in Python using Jupyter notebook. We used the following packages available in Python to get the results:

1. pandas
2. NumPy
3. scikit-learn
4. community
5. NetworkX
6. Matplotlib
7. Seaborn

Result Discussions

Our analysis reveals a consistent structure in the way these nine Community Areas relate to one another. In the normalized feature space, the three affluent North Side neighborhoods cluster tightly together sharing high income, high educational attainment, low poverty and unemployment, limited reliance on SNAP, and strong broadband access. On the other hand of extreme similarity, Englewood and West Englewood are similar to each other in the aspects of very high poverty, unemployment, dependance on SNAP, and lower levels of education and broadband access. In the middle, we have Northwest neighborhood and South Lawndale, which have mixed profiles and intermediate values on many of the data variables.

In the cosine graph, the three North Side neighborhoods form a tightly connected group, Englewood and West Englewood form a strongly linked pair that is weakly connected to the rest of the network, and the Northwest areas and South Lawndale acted as bridges between different parts of the structure. High income neighborhoods also have high degrees and high eigenvector centrality, reflecting their position in a core of mutually similar areas. Englewood and West Englewood showed low centrality and is effectively cut off. As for the Northwest areas, closeness centrality is relatively high, which represents their role as a bridge in the structured network.

Community detection and clustering provide further evidence of our neighborhood hierarchy. Louvain community detection on the main graph also consistently identifies a group of affluent North Side neighborhoods, a pair containing Englewood and West Englewood, and an intermediate community that includes the Northwest areas along with South Lawndale. Additionally, when we apply k means clustering to the same feature space and view the result in PCA coordinates, we can see that Englewood and West Englewood form one cluster, the high- and middle-income North and Northwest neighborhoods form another, and South Lawndale becomes its own separate cluster.

Across different network constructions, we can see that the following patterns:

- Affluent North Side areas almost always remain in the same cluster and near the center of the network.
- Englewood and West Englewood almost always form a tight pair and remain relatively isolated from the rest of the community areas.

Assortativity measurement graph indicates a strong positive assortativity between income and education, which also means that neighborhoods that are connected in the network tend to have similar socioeconomic tiers. We can also observe assortativity by race, confirming that racial composition contributes to the way Community Areas group together.

A lot of our models aligned with our expectations, but the most interesting and unexpected pattern was the isolated cluster of South Lawndale in the k means clustering. South Lawndale was picked as one of the medium-income/mix profiles of the neighborhood. Additionally, it also appears in an intermediate community that links the North and the South Side in our network models.

We were able to successfully model Community Areas as nodes in a similarity network based on a dataset of specific socioeconomic variables, determining segregation and clustering. Utilizing multiple methods and network constructions also helped us confirm that the patterns were robust. However, we only worked with nine Community Areas, therefore our networks only have nine nodes, so some statistics may more be illustrative than rigorous, and any small changes in a single edge can have a noticeable impact on the overall network.

For our evaluation process, we made sure that the code, tables, and visualizations are consistent with one another. We verified that the edges in the graph correspond to the strongest entries in the cosine similiarity matrix, the community labels in the CSV files matches the community plots, and the cluster labels used in the PCA figure are the same as those produced by k means. The project could be evaluated externally by comparing our clusters and similarity patterns with additional public resources or any other existing case studies on Chicago’s geography of inequality, to see how closely our project aligns with any established papers.

In future work, we could extend this project into several directions:

1. Scale from 9 Community Areas to all Community Areas
2. Track changes of each data variable over time (2018-2023)

3. Add additional economic indicators such as inflation, cost of living, and analyze how they correlate with income, rent, unemployment, and reliance on assistance
4. Combine these concepts by building a longitudinal network that indicates how similarities between neighborhoods evolve across years and how those changes relate to broader economic and policy shifts

Code, Data & AI Tools

Code, Data & Output: https://github.com/aungkham-naung/ONSA-Final

Census Raw Data: ONSA Raw Data

Chat GPT transcript: **https://chatgpt.com/share/6938c846-9a1c-800f-ac3c-398bf56bfd12**